Appendix A

IRT Model Comparisons

For each dataset, we fitted each of the four standard psychometric models (1PL-4PL) and compared the fits using Akaike and Bayesian Information Criteria (AIC and BIC) to select models with the appropriate level of complexity (i.e. number of parameters) justified by the data. Tables A1-A4 show the results of these model comparisons.

English Comprehension. Table A1 shows the model comparison for the 2394 children in the English comprehension data. Both AIC and BIC prefer the 2PL model over either the simpler Rasch (1PL) model or the more complex 3PL or 4PL models. Note that the 4PL model converged very quickly, yet with a worse overall fit than the other models: this remained the case even after setting a much lower tolerance for convergence. Phil Chalmers, the author of the mirt R package, indicates that the 4PL model has too many parameters (and thus flexibility) unless the dataset is in excess of 5000 participants.

Table A1

Comparison of 1PL-4PL models for English comprehension.

Model	AIC	BIC	logLik	df
Rasch	718,890.44	721,185.38	-359,048.22	NA
2PL	708,467.70	713,046.03	-353,441.85	395.00
3PL	708,909.35	715,776.85	-353,266.68	396.00
4PL	709,382.34	718,539.00	-353,107.17	396.00

Spanish Comprehension. Table A2 shows the model comparison for the 759 children in the Spanish comprehension data. The 2PL model is preferred over both the Rasch (1PL) model and the 3PL model. As for the English comprehension dataset, the

4PL again converged quickly, but with worse fit than the 3PL, likely due to the small size of the dataset relative to the number of parameters.

Table A2

Comparison of 1PL-4PL models for Spanish comprehension.

Model	AIC	BIC	logLik	df
Rasch	238,610.98	240,598.10	-118,876.49	NA
2PL	236,007.27	239,972.27	-117,147.64	427.00
3PL	236,590.80	242,538.29	-117,011.40	428.00
4PL	254,849.39	262,779.38	-125,712.70	428.00

English Production. Table A3 shows the model comparison for the 7633 children in the English production data. For both model selection criteria, smaller values are better. The 2PL model is preferred over the Rasch (1PL) model by both AIC and BIC, indicating that the 2PL's addition of a discrimination parameter (slope) for each item is justified. The 3PL model is preferred over the 2PL by AIC, but not by the more conservative BIC criterion. AIC again prefers the more complex 4PL model over the simpler 3PL model, but according to BIC the 2PL model has the appropriate level of complexity for the dataset.

Table A3

Comparison of 1PL-4PL models for English production.

Model	AIC	BIC	logLik	df
Rasch	2,542,533.33	2,547,259.63	-1,270,585.66	NA
2PL	2,475,247.30	2,484,686.02	-1,236,263.65	679.00
3PL	2,472,824.63	2,486,961.89	-1,234,375.31	677.00
4PL	2,466,697.98	2,485,547.67	-1,230,632.99	679.00

Spanish Production. Table A4 shows the model comparison for the 1610 children in the Spanish production data. The 2PL model has the lowest AIC value, as well as the lowest BIC value, indicating that it is of the appropriate complexity for this dataset.

Table A4

Comparison of 1PL-4PL models for Spanish production.

Model	AIC	BIC	logLik	df
Rasch	576,073.90	579,740.40	-287,355.95	NA
2PL	559,215.47	566,537.69	-278,247.73	679.00
3PL	560,716.88	571,700.22	-278,318.44	680.00
4PL	612,960.38	627,604.84	-303,760.19	680.00

Appendix B

Ill-fitting Items

Below we show the full list of items with ill fit (Stone's $\chi^2 *_{df}$, $p < .001$) in the 2PL model for Spanish production (38 items) and for English production (142 items).

Table B1

Ill-fitting Spanish production items

¡am!	éste	mesa	pies	televisión
a	gritar	mío	quién	tengo manita
adíos/byebye	guaguá	mirar	recámara	tortilla
árbol	huevo	nana	saber	tú
atole	jugar	no	shhh	y
boca	las	o	suya	ya
caja	manos arriba	oír	taco	
caliente	me	pelo	te (pronouns)	

Table B2

Ill-fitting English production items

all	corn	hand	muffin	school	this
all gone	couch	hear	my	see	throw
baby	cow	helicopter	nice	sheep	tickle
that	daddy*	hello	noodles	shoe	to
ball	did/did ya	hi	nose	sing	tooth
bathroom	dirty	home	not	sister	towel
bathtub	don't	hungry	orange (adj)	sled	trash
beans	donkey	I	ouch	sleep	tree
bee	donut	is	out	sleepy	tummy
blanket	dress (noun)	it	owie/boo boo	snow	uh oh
boots	ear	jacket	pancake	sock	vroom
bottle	eye	jar	peas	spoon	water (not bev)
brother	fish (food)	juice	pen	stick	what
brown	french fries	keys	penis*	stop	which
bump	friend	look	pickle	story	white
by	garage	love	pizza	stroller	who
bye	get	man	popcorn	stuck	woof woof
child	girl	me	potty	sweater	yes
choo choo	good	meat	purse	swing (verb)	yogurt
church*	goose	mine	raisin	table	you
coat	grrr	mommy*	read	teddybear	yucky
coffee	gum	money	red	thank you	yum yum
coke	hair	motorcycle	ride	there	babysitter's
cold	hamburger	mouth	rooster		name

Appendix C

Simulated CATs

The following sections show the results of the simulated CATs using American English and Mexican Spanish data, conducted for both production (combined WG and WS data) and comprehension (WG data).

Fixed-length CAT Simulations

Following Makransky et al. (2016), we ran a series of fixed-length CAT simulations and compared children's estimated ability (θ) from these CATs to children's estimated ability from the full CDI. The results are shown in Tables C1-C4 for English comprehension (C1), Spanish comprehension (C2), English production (C3), and Spanish production (C4). The results were quite good even for 25- and 50-item tests, but note that we added a comparison to tests of randomly-selected questions (per subject), and found that ability estimates from these tests were also strongly correlated with thetas from the full CDI. The mean standard error of the random tests showed more of a difference, especially for shorter tests (e.g., <100 items).

Table C1

Fixed-length CAT simulations compared to full CDI for English comprehension

Test Length	r vs. full CDI	Mean SE	Reliability	r Random vs. full CDI	Random SEM
25	0.980	0.175	0.969	0.949	0.296
50	0.990	0.137	0.981	0.975	0.220
75	0.994	0.120	0.986	0.982	0.184
100	0.996	0.110	0.988	0.989	0.161
200	0.999	0.093	0.991	0.996	0.118
300	1.000	0.087	0.992	0.999	0.097

Table C2

Fixed-length CAT simulations compared to full CDI for Spanish comprehension

Test Length	r vs. full CDI	Mean SE	Reliability	r Random vs. full CDI	Random SEM
25	0.970	0.192	0.963	0.935	0.317
50	0.983	0.152	0.977	0.962	0.238
75	0.989	0.135	0.982	0.975	0.202
100	0.992	0.124	0.985	0.982	0.180
200	0.998	0.105	0.989	0.994	0.133
300	1.000	0.098	0.990	0.998	0.112

Table C3

Fixed-length CAT simulations compared to full CDI for English production.

Test Length	r vs. full CDI	Mean SE	Reliability	r Random vs. full CDI	Random SEM
25	0.990	0.157	0.975	0.953	0.308
50	0.995	0.126	0.984	0.971	0.242
75	0.996	0.113	0.987	0.980	0.209
100	0.997	0.106	0.989	0.985	0.188
200	0.999	0.095	0.991	0.993	0.143
300	0.999	0.091	0.992	0.996	0.122
400	1.000	0.088	0.992	0.998	0.108

Table C4

Fixed-length CAT simulations compared to full CDI for Spanish production.

Test Length	r vs. full CDI	Mean SE	Reliability	r Random vs. full CDI	Random SEM
25	0.987	0.182	0.967	0.934	0.329
50	0.992	0.151	0.977	0.958	0.265
75	0.994	0.138	0.981	0.972	0.233
100	0.996	0.130	0.983	0.978	0.213
200	0.998	0.117	0.986	0.991	0.166
300	0.999	0.111	0.988	0.994	0.144
400	0.999	0.109	0.988	0.997	0.129

Early-stopping CAT Simulations

For each participant in each dataset, we simulated CATs using a maximum of 25, 50, 75, 100, 200, 300, or (for production datasets only) 400 items, with early termination if the estimated standard error of measurement (SEM) reached 0.10 for a given subject. For each of these simulations, we report 1) the mean number of items used, 2) the correlation of ability scores (θ) estimated from the CAT with IRT-estimated ability scores applied to the full CDI (*r with full CDI*), and 4) the mean estimated SEM achieved by the CATs, 5) the reliability of the ability estimates from the CAT compared to the ability estimated by the full CDI, and 6) how many items were never selected (*Unused Items*). The results are shown in Tables C5-C9.

Table C5

Early-stopping CAT simulations for English comprehension.

Max. Items	Mean Items	r with full CDI	Mean SE	Reliability	Unused Items
25	25.0	0.980	0.175	0.969	235
50	50.0	0.990	0.137	0.981	145
75	73.5	0.994	0.121	0.985	76
100	84.5	0.995	0.116	0.986	52
200	107.2	0.996	0.113	0.987	0
300	125.0	0.996	0.112	0.987	0

Table C6

Early-stopping CAT simulations for Spanish comprehension.

Max. Items	Mean Items	r with full CDI	Mean SE	Reliability	Unused Items
25	25.0	0.970	0.192	0.963	276
50	50.0	0.983	0.152	0.977	172
75	70.4	0.988	0.137	0.981	114
100	81.6	0.990	0.132	0.983	83
200	113.1	0.993	0.125	0.984	2
300	138.3	0.994	0.123	0.985	0

Table C7

Early-stopping CAT simulations for English production.

Max. Items	Mean Items	r with full CDI	Mean SE	Reliability	Unused Items
25	24.7	0.990	0.158	0.975	468
50	39.5	0.993	0.136	0.981	405
75	49.9	0.994	0.131	0.983	375
100	58.7	0.995	0.128	0.983	344
200	88.9	0.995	0.126	0.984	183
300	116.4	0.995	0.125	0.984	87
400	143.1	0.995	0.124	0.985	15

Table C8

Early-stopping CAT simulations for Spanish production.

Max. Items	Mean Items	r with full CDI	Mean SE	Reliability	Unused Items
25	25.0	0.987	0.182	0.967	496
50	40.3	0.991	0.160	0.974	403
75	51.6	0.992	0.154	0.976	352
100	61.9	0.993	0.150	0.977	308
200	98.6	0.994	0.145	0.979	176
300	133.0	0.994	0.143	0.980	80
400	166.3	0.994	0.142	0.980	12

CAT Performance Across Age Groups

For the preferred settings, it is desirable to know if the CAT can be expected to perform well across the intended age range. Thus, for each dataset we examined correlations between the simulated preferred CAT's ability estimates to the IRT-estimated ability from the full CDI.

English Comprehension. Table C9 shows correlations between ability estimates from the full CDI compared to the estimated ability from the preferred CAT by age group (126 8-9 month-olds, 233 10-11 mos, 909 12-13 mos, 233 14-15 mos, and 893 16-18 mos.

Spanish Comprehension. Table C10 shows correlations between ability estimates from the full CDI compared to ability from the preferred CAT by age group (104 8-9 month-olds, 138 10-11 mos, 172 12-13 mos, 137 14-15 mos, and 208 16-18 mos.

English Production. Table C11 shows correlations between ability from the full CDI compared to ability from the preferred CAT split by age (1035 12-15 month-olds, 2156

Table C9

Correlation between the preferred CAT's ability estimates and the full CDI for English comprehension by age group.

Scoring / Start Item	[8,10)	[10,12)	[12,14)	[14,16)	[16,18]
ML / MI	0.963	0.976	0.974	0.975	0.975
MAP / MI	0.983	0.977	0.974	0.976	0.977
ML / age-based	0.965	0.975	0.974	0.974	0.975
MAP / age-based	0.982	0.976	0.973	0.975	0.977

15-18 mos, 1128 18-21 mos, 665 21-24 mos, 1095 24-27 mos, 1190 27-30 mos, 322 30-33 mos, and 42 33-36 mos).

Spanish Production. Table C12 shows correlations between ability from the full CDI compared to ability from the preferred CAT split by age (248 12-15 month-olds, 318 15-18 mos, 307 18-21 mos, 227 21-24 mos, 219 24-27 mos, and 291 27-30 mos).

Table C10

Correlation between the preferred CAT's ability estimates and the full CDI for Spanish comprehension by age group.

Scoring / Start Item	[8,10)	[10,12)	[12,14)	[14,16)	[16,18]
ML / MI	0.958	0.97	0.962	0.978	0.961
MAP / MI	0.964	0.978	0.963	0.978	0.961
ML / age-based	0.957	0.971	0.963	0.979	0.961
MAP / age-based	0.965	0.978	0.963	0.979	0.961

Table C11

Correlation between the preferred CAT's ability estimates and the full CDI for English production by age group.

Scoring / Start Item	[12,15)	[15,18)	[18,21)	[21,24)	[24,27)	[27,30)	[30,33)	[33,36]
ML / MI	0.966	0.975	0.978	0.984	0.977	0.971	0.943	0.934
MAP / MI	0.985	0.979	0.98	0.985	0.977	0.971	0.944	0.939
ML / age-based	0.966	0.976	0.978	0.985	0.977	0.973	0.955	0.919
MAP / age-based	0.985	0.98	0.979	0.985	0.977	0.973	0.957	0.94

Table C12

Correlation between the preferred CAT's ability estimates and the full CDI for Spanish production by age group.

Scoring / Start Item	[12,15)	[15,18)	[18,21)	[21,24)	[24,27)	[27,30]
ML / MI	0.912	0.971	0.975	0.974	0.976	0.969
MAP / MI	0.972	0.981	0.978	0.977	0.978	0.97
ML / age-based	0.918	0.971	0.974	0.974	0.977	0.97
MAP / age-based	0.973	0.981	0.977	0.977	0.979	0.972

Appendix D

Ability vs. Age

For participants in the validation study, Figure D1 shows the relationships between children's age and 1) their total vocabulary from the full CDI:WS (left), 2) their full CDI ability (middle), and 3) their ability measured from the CDI-CAT (right). These relationships are all of approximately the same strength, and show the expected female advantage.

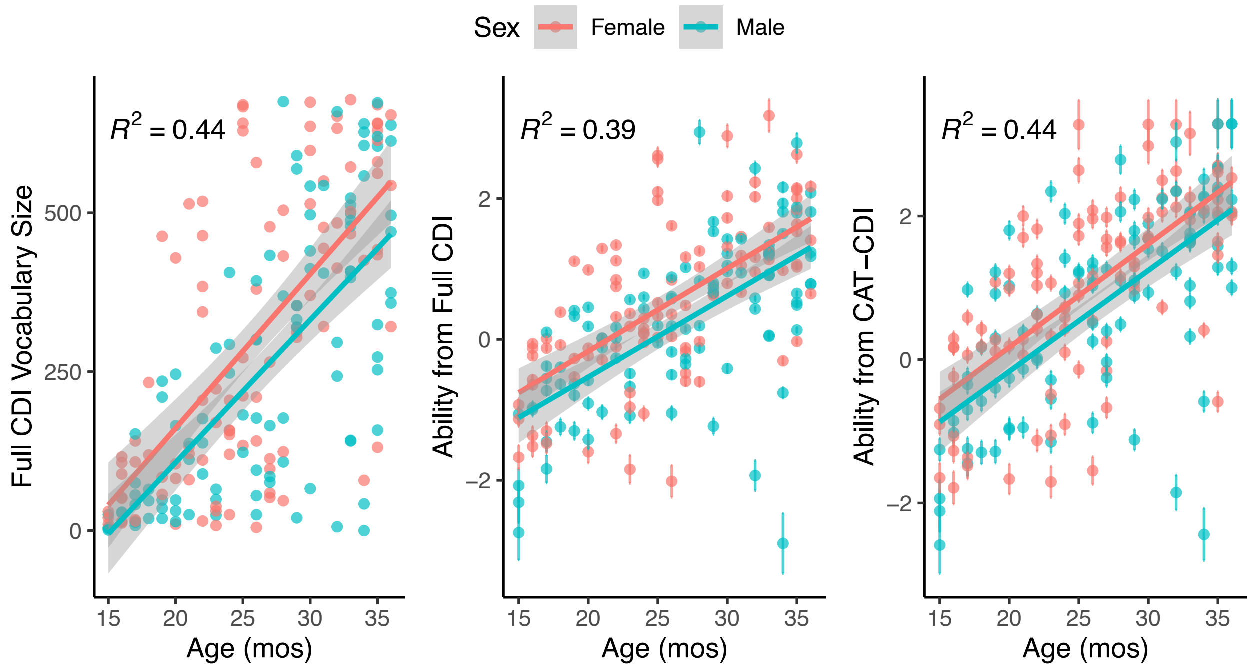

Figure D1. Children's age vs. their full CDI vocabulary size, full CDI ability, and CAT-estimated ability, by sex of child.

Appendix E

Standard Error of Measurement by Ability

Finally, we examine the standard error of measurement (SEM) for children of varying ability using the preferred CAT settings with age-based starting items. The SEM theoretically varies with ability, as the CDI is most informative of children's ability for children who are of closer to average ability. This is because there are more items of intermediate difficulty than very difficult or very easy items, and so the test is better calibrated to distinguish children of intermediate ability than children at high/low ranges of ability. Figure E1 shows SEM vs. estimated ability (theta) for each child in the Wordbank data using the preferred CDI-CAT settings. The CDI-CATs show fairly minimal SEM for children with ability $-1 \leq \theta \leq 2$, meaning that the CDI-CATs will give a quite precise estimate of ability for $\sim 82\%$ of children. For children with estimated $\theta < -2.5$ or $\theta > 3$ (i.e. $< 2\%$ of the population) the SEM starts to rise, and it may be advisable to administer a full CDI if it is important to get a precise estimate of the child's language ability.

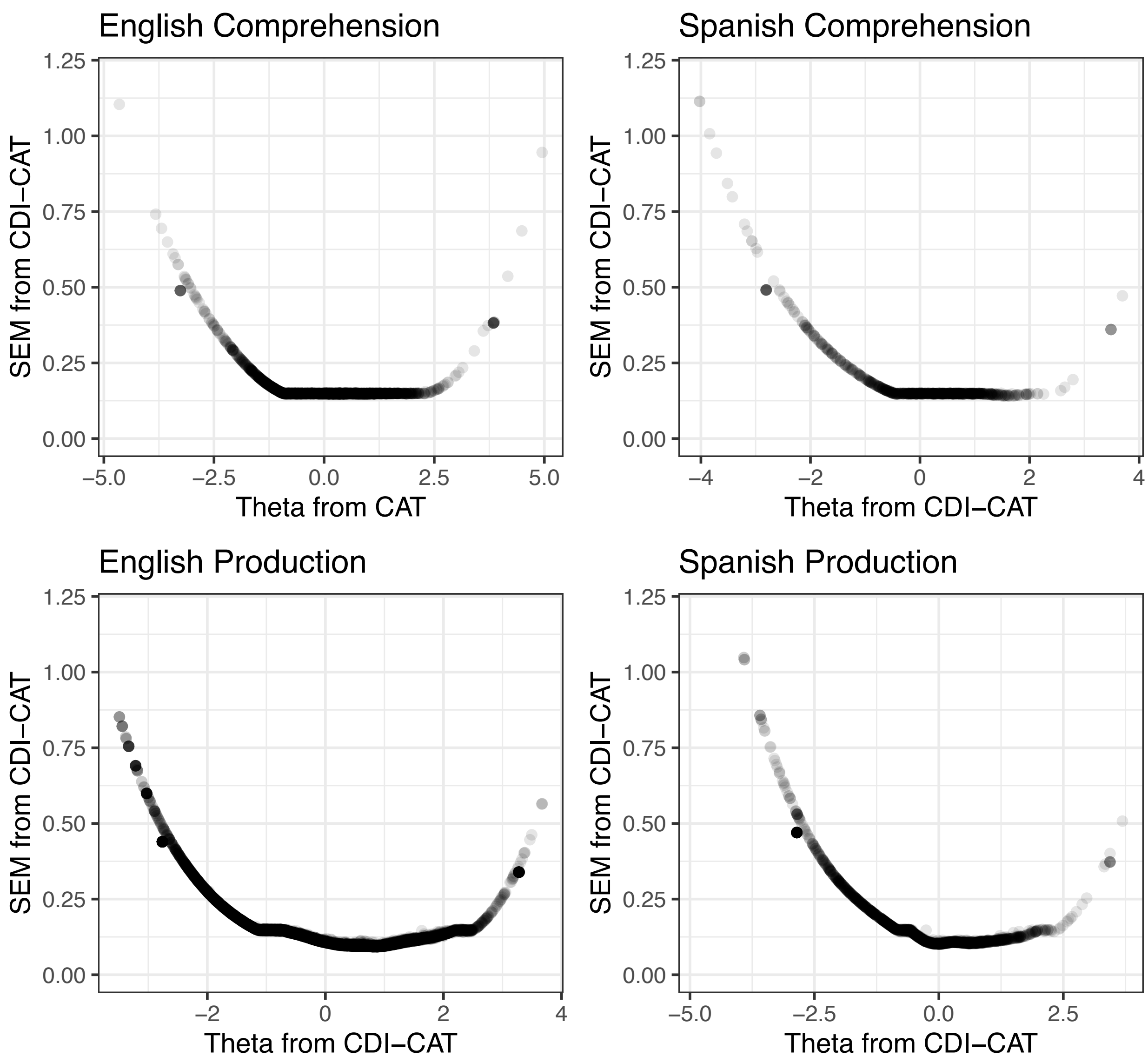

Figure E1. Standard Error of Measurement (SEM) vs. children's language ability (theta) using the preferred CDI-CAT settings, with ML-based estimation and age-based starting items).

Appendix F

Comparison of CDI-CAT and CDI Short Forms

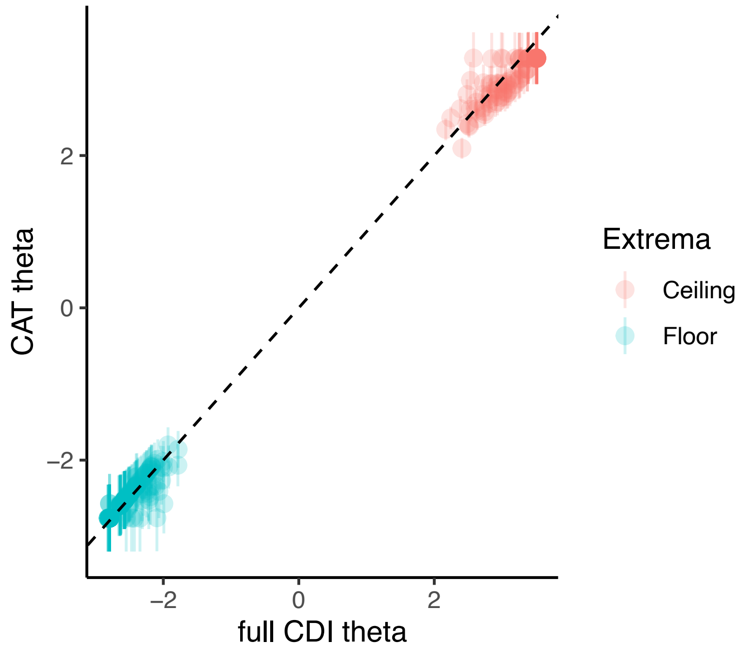

Figure F1. Children's CDI-CAT estimated language ability (theta) vs. IRT-estimated ability from the full CDI. The only children shown are the 308 who would have gotten a floor or ceiling score on one or more 100-item short-form CDIs. On the CDI-CAT, most of these children no longer show a ceiling or floor effect, and their CAT-estimated ability is strongly related to the IRT-estimated ability from the full CDI.

Finally, we investigated whether the CDI-CAT would mitigate floor and ceiling effects that would be present if participants in the Wordbank English production dataset (N=7633) were given existing 100-item short-form CDIs from Fenson et al. (2000). In our dataset, 213 children would receive a floor score on one or both of the short-form production CDI:WS forms (i.e., be reported by their caregiver to produce 0 of the 100 items), and 95 children would receive a ceiling score on one or both of the short-form production CDIs (i.e., be reported to produce all of the 100 items). Figure F1 shows the estimated ability for these 308 children as estimated by the production CDI-CAT with preferred settings (MAP-based estimation with age-based start items) vs. their

IRT-estimated ability from the full CDI. For these children who would have scored at floor or ceiling on the short-form CDIs, only 18 scored at floor on the CDI-CAT, and 1 scored at ceiling on the CDI-CAT. Overall, the CDI-CAT thetas vs. the full CDI thetas for these children were strongly related ($r = .999$, $t(306) = 337.9$, $p < .001$). Thus, we can conclude that the CDI-CAT will mitigate ceiling and floor effects for many children (289 out of 308 in this sample) that would be found if the 100-item short-form CDIs were used – and with at most half the number of tested items.